

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PTE 316 — Past Questions & Materials
by Renance

Course: PTE 316
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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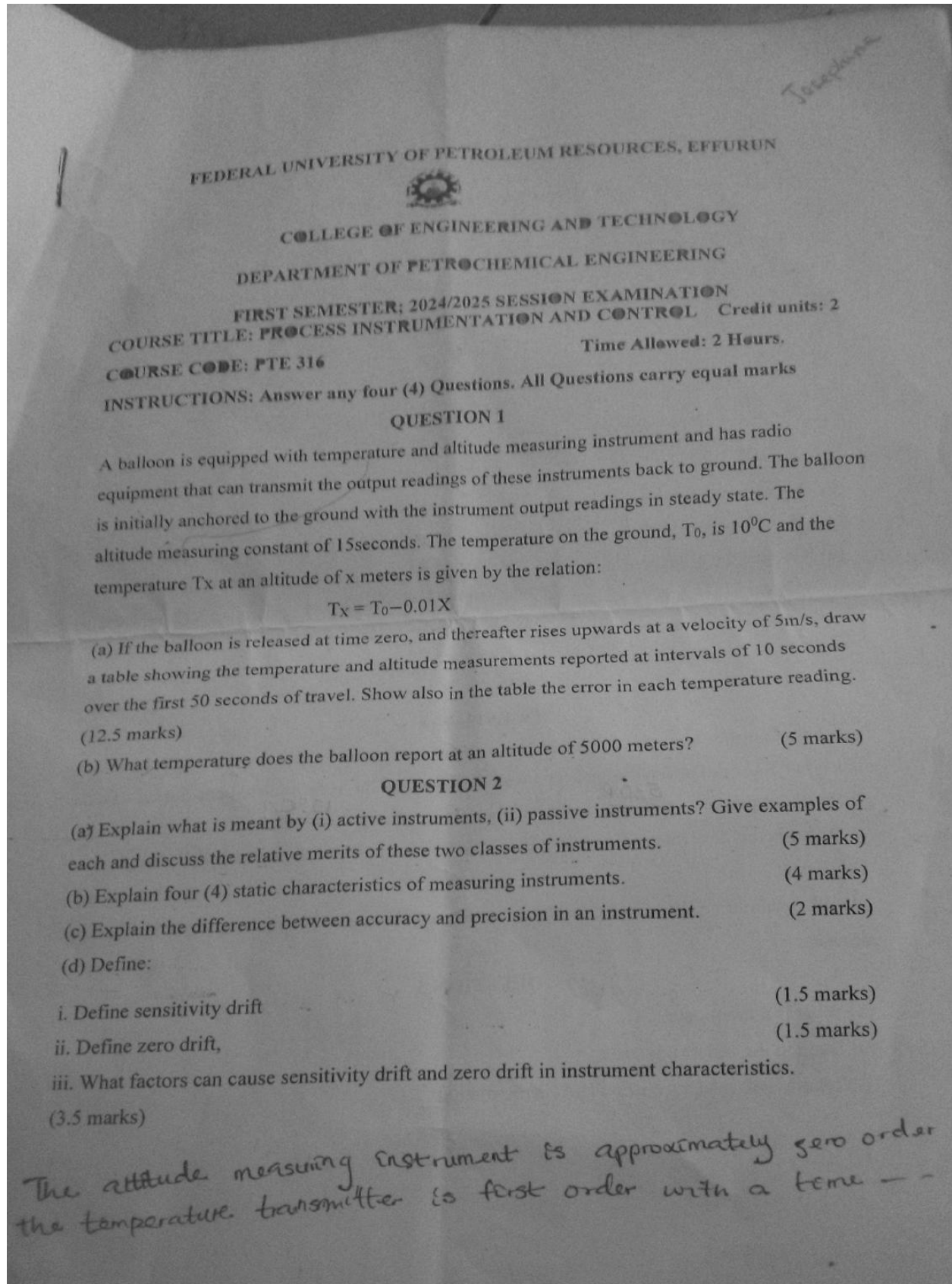
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Paper 1 of 1 — PTE 316 - 2024/2025 - Second Semester - PROCESS INSTRUMENTATION AND CONTROL

Level 200 • Second Semester • PastQuestion

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QUESTION 3 ✓

- (a) In a temperature measuring system the transfer function is $3.2 \text{ MV/k} \pm 2.1\%$ and the accuracy of the transmitter is $\pm 1.7\%$. What is the system accuracy? (2.5 marks)
- (b) The following resistance values of a platinum resistance thermometer were measured at a range of temperatures. Determine the measurement sensitivity of the instrument in $\text{ohms}/^\circ\text{C}$.

Resistance (Ω)	Temperature ($^\circ\text{C}$)
307	200
314	230
321	260
328	390

- (c) The pressure in a system has a range from 0 to 75 kpa. What is the current equivalent of 27 kpa, if the transducer output range is from 4 to 20 mA? (2.5 marks)
- (d) Describe the principles of operation of the following volumetric flow meters: (2.5 marks)
- Oval gear (2.5 marks)
 - Rotary piston (2.5 marks)
 - Impeller (2.5 marks)
 - Gas (2.5 marks)

QUESTION 4 ✓

- (a) State four problems and two applications of the turbine flow meter. (6.5 marks)
- (b) What is the radiation constant for a wall 5.80m by 2.75m , if the radiated heat loss is 410KW , the wall temperature is 530°R , and the ambient temperature is 12°C ? (3.5 marks)
- (c) What is the flow rate in litres per second through a pipe 32cm in diameter, if the average velocity is 2.1m/s ? (3.5 marks)
- (d) What should be taken into considerations when installing and calibrating pressure sensing devices. (4 marks)

QUESTION 5 ✓

- (a) What is a thermostat? (2 marks)
- (b) Explain the following terms: (6 marks)
- Conduction, (ii) Convection and (iii) Radiation

(c) State four advantages and four disadvantages of mercury as a thermometric liquid? (7 marks)

(d) Mention two thermometric liquids that can be used at temperature below 110°C . (2.5 marks)

QUESTION 6 ✓

6.(a) Explain the principle and application of the following temperature measuring devices:

i. Bimetallic thermometers and thermostats (3.5 marks)

ii. Liquid-in-glass thermometers (3.5 marks)

iii. Gas filled thermometers (3.5 marks)

iv. Vapour pressure thermometers (3.5 marks)

(b) A straight bimetallic strip 25cm long and 1.4mm thick is made of copper and invar, and has one end fixed. How much will the free end of a strip deflect if the temperatures changes 18°C ?

Co-efficient of linear expansion of Copper is 16.9×10^{-6} and Invar is 1.2×10^{-6} respectively.

(3.5 marks)